

Vadaanya Talent Test 2026

System Architecture & End-to-End Implementation Guide

Two-Step Entry Gate · Database Drafts · Aadhaar-Safe Encryption · Hall Tickets · Pure Analytics Dashboard · Build Roadmap

8,000
Students (Classes 9 and 10)

2
Districts: Anantapur & Sri Sathya Sai

64
Mandals (32 per district)

384+
Eligible Rural Schools

Dec 15
Exam Day, 2026

Document Version	2.0.0 (Updated with Step-0 Gate, DB Drafts, Pure Analytics & V26 Format)
Build Target Completion	October 16, 2026
Registration Window	November 1, 2026 – November 30, 2026
Exam Date	December 15, 2026
Live Portals	www.vadaanya.org (Public) · admin.vadaanya.org (Analytics Dashboard)
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How to read this guide: Every section opens with a plain-language summary for project leadership, followed by in-depth technical specifications, SQL schemas, sequence diagrams, and edge-case countermeasures.

Two kinds of "phase" in this document:

Operational Stages: Describe what happens to students and schools over calendar time: Registration (Nov 1–30), Logistics Batch (Dec 1–6), Hall Tickets (Dec 7–15), and Exam Day (Dec 15).

Build Phases: Describe the engineering sprints from Oct 4 to Oct 16 to build and test the software.

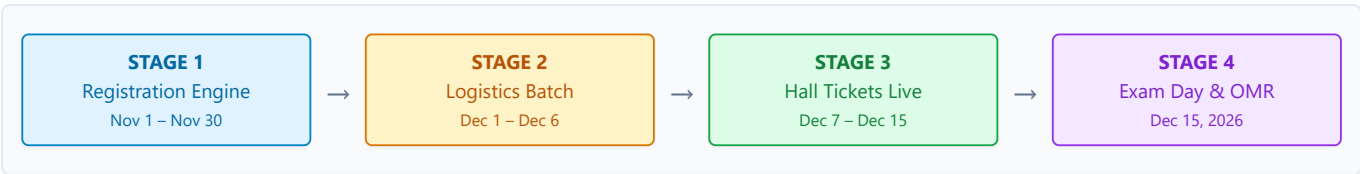
1. Project at a Glance

Vadaanya Talent Test 2026 transitions a five-year-old, paper-based rural talent test onto a resilient, automated cloud platform. It registers 8,000 government school students across Anantapur and Sri Sathya Sai districts, manages exam center allocations, generates verifiable hall tickets, and prepares OMR-ready data for exam day.

1.1 What is Changing

Before (Manual Paper Process)	After (Vadaanya 2026 Cloud Platform)
Physical paper forms filled by hand; data transcription errors.	Mobile-first web form with instant Verhoeff validation and database auto-save.
Incomplete forms discarded or lost due to intermittent connectivity.	Database-Backed Drafts: Uncompleted forms saved in DB as <code>PENDING</code> and auto-resumed on Aadhaar re-entry.
Manual alphabetical sorting across mandals in spreadsheets.	Automated spatial and mandal clustering with gender-segregated exam center assignments.
Hand-delivered hall tickets requiring weeks of logistical delay.	Instant PDF hall tickets with QR verification, WhatsApp/SMS links, and one-click Headmaster bulk downloads.
Zero visibility into registration momentum or drop-off friction.	Dedicated Admin Analytics Dashboard tracking funnel drop-offs, district quotas, and low-registration school red flags.

1.2 Operational Lifecycle in Four Stages



Stage	Dates	What Happens	Key Deliverable
1. Registration	Nov 1 – Nov 30	Students or teachers register via mobile. Step 0 gate checks duplicates and resumes drafts. Data auto-saves to database. Quotas enforced at 4,000 each.	Registration Number (e.g. <code>V26-ATP-B0016</code>), Digital Receipt, WhatsApp Confirmation.
2. Logistics Batch	Dec 1 – Dec 6	Registration is frozen. Proximity algorithm clusters students by mandal/village, separates boys and girls into centers, and assigns seats.	Roll numbers, room & bench allocations, center master files.
3. Hall Tickets	Dec 7 – Dec 15	Tickets go live 1 week before exam. Mass WhatsApp broadcast triggered. Headmasters download single stitched batch PDFs.	Bilingual A4 PDF with high-contrast QR code for gate scanning.
4. Exam Day & OMR	Dec 15	Invigilators verify students using the Spot-Verification search tool. Admins export pre-formatted OMR Excel file.	Spot-verification lookup, OMR Scanner Master Excel (.xlsx).

2. Rural Ground Realities & Technical Responses

The platform is engineered around the actual constraints of rural Andhra Pradesh, ensuring accessibility, data integrity, and high completion rates.

2.1 Stakeholder Personas & Realities

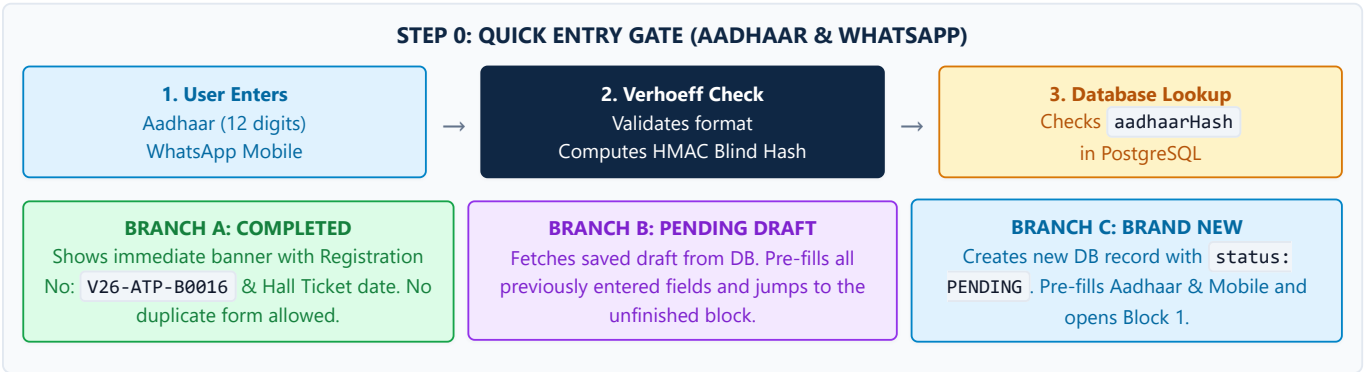
Persona	Ground Reality	System Requirement
Rural Student (Class 9 & 10)	Lacks personal smartphone, computer, or email address. Official identifier is their 12-digit Aadhaar. Language is Telugu and simple English.	Never require email. Use Aadhaar as sole identity and mobile number purely for notifications. Bilingual Telugu/English interface.
School POC (Headmaster / Math Teacher)	Registers 20 to 50 students in one sitting using one school PC, mobile phone, or local MeeSeva center. One phone number repeats across dozens of students.	Allow 1:N phone-to-student mapping. Provide "Quick Add Next Student" mode. Provide one-click School Batch PDF download.
Field Volunteer (50-Member Campaign Team)	Visits remote schools with tablets/phones after District Collector launch. Network drops between 2G, 3G, and 4G.	Lightweight Next.js payload (<80KB gzip). Database-backed block auto-saving ensures partially filled forms are never lost.

2.2 Problem & Technical Countermeasure Matrix

Ground Hurdle	Why Traditional Systems Fail	Vadaanya 2026 Architectural Solution
Phone Number Collisions	Using phone numbers as a unique constraint crashes when a teacher registers 30 students.	Aadhaar is the sole unique identifier. Mobile numbers are treated solely as delivery routes, permitting 1:N mapping without conflict.
Dropped Forms / Power Cuts	Unstable internet causes students to lose half-filled forms, leading to abandonment.	Database-Backed Auto-Save: Each block entered is immediately saved to PostgreSQL under <code>status: PENDING</code> . Re-entering Aadhaar auto-restores data on any device.
Duplicate Registrations	Students forget they were registered by a teacher and try again, creating duplicates.	Step 0 Gate Check: Entering Aadhaar immediately tells them: <i>"You have already registered. Your Reg No is V26-ATP-B0016"</i> without loading the form.
Lost Registration Slips	Students lose paper slips or delete SMS, unable to retrieve hall tickets.	Dual-Key Recovery: Hall tickets can be retrieved with Registration Number OR <code>Aadhaar + Mobile Number</code> .
Teacher Bulk Entry Fatigue	Re-typing District, Mandal, and School 30 times causes extreme fatigue and typos.	"Quick Add Next Student" Mode: Retains District, Mandal, and School in form state, resetting only student-specific fields.

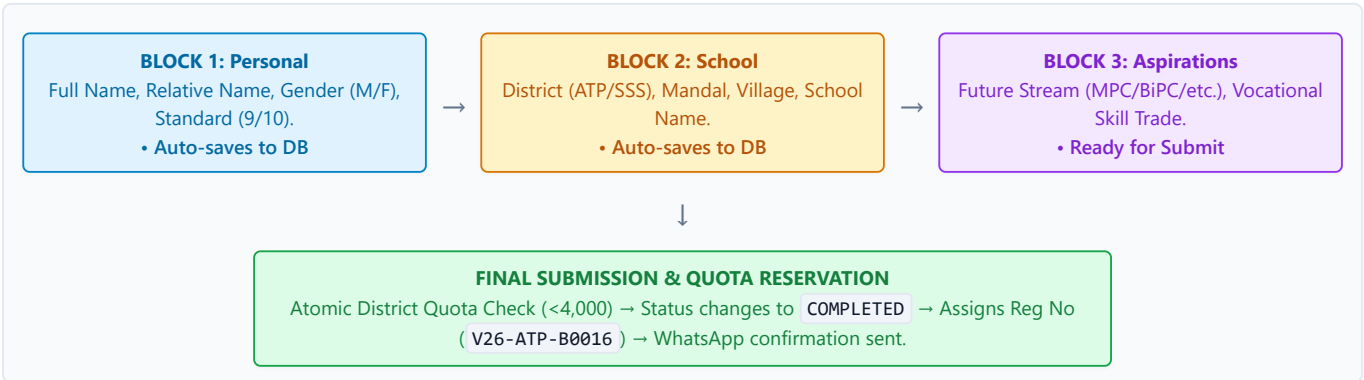
2.3 The Two-Step Registration Architecture

To eliminate friction, the registration process begins with an **Initial Entry Gate (Step 0)** before opening the structured form blocks.



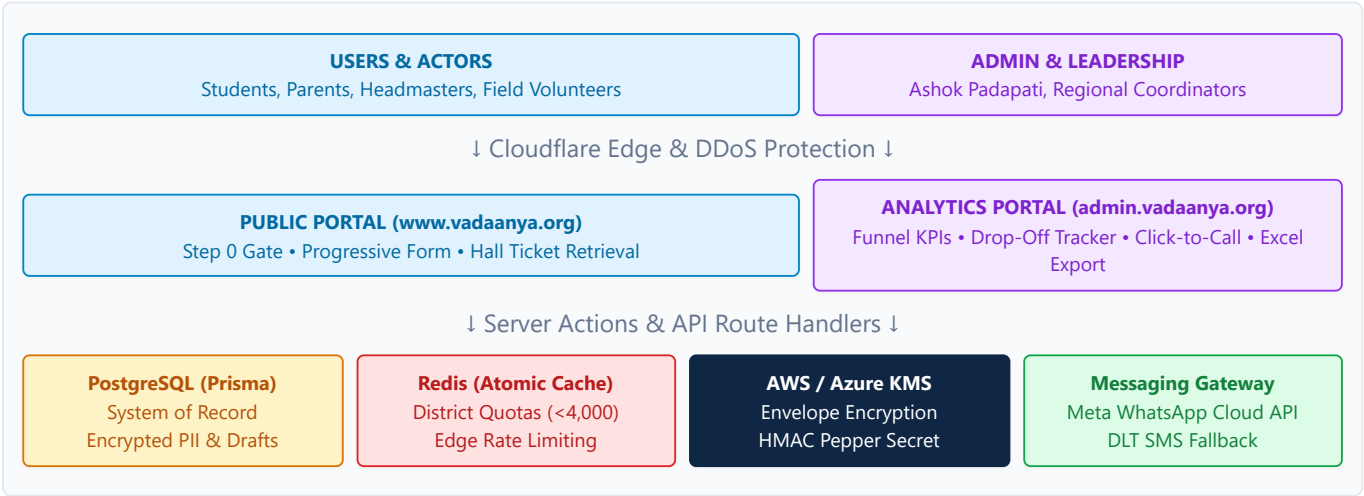
2.4 Progressive Block-by-Block Form Split & Auto-Save

Rather than presenting a monolithic 12-field form, fields are grouped into three progressive blocks. Each block automatically commits its payload to PostgreSQL via `PATCH /api/registration/draft`.



3. System Architecture

The platform is deployed as a single Next.js 16 application serving two completely isolated entry domains: `www.vadaanya.org` for public registration and `admin.vadaanya.org` for internal analytics and monitoring.



3.1 Core Technology Stack

Layer	Choice	Role in Vadaanya 2026
Frontend	Next.js 16 (App Router), React 19, TypeScript, Vanilla CSS	Lightweight, fast-loading bundle (<80KB gzip) designed for rural 2G/3G mobile devices.
Backend	Next.js Server Actions & Route Handlers	Server-side validation, Verhoeff checksum verification, draft auto-saving, quota locks.
Database	PostgreSQL with Prisma ORM	System of record; atomic seat increments, unique blind index constraints, draft status management.
Security / KMS	AWS KMS or Azure Key Vault	AES-256-GCM envelope encryption keys and HMAC pepper secret storage.
Document Engine	<code>@react-pdf/renderer</code> / <code>puppeteer-core</code>	Generates dynamic bilingual A4 Hall Ticket PDFs and concatenated School Batch PDFs.
Messaging	Meta WhatsApp Cloud API + DLT Indian SMS	Instant confirmation, hall ticket download broadcast, and 48-hour exam countdown reminder.

4. Aadhaar Data Protection & Zero-Leakage Architecture

The Aadhaar number is sensitive Personally Identifiable Information (PII) governed by strict UIDAI regulations. **Plain-text Aadhaar numbers must NEVER exist in the database, cache, or logs.**

4.1 The Four Security Protection Layers

Layer	Mechanism	Database Field	Visibility & Access
1. Deterministic Blind Index	HMAC-SHA256(normalised_aadhaar, KMS_PEPPER) . Pepper is stored in Key Vault, never in code. Enforces SQL UNIQUE constraint for instant \$O(1)\$ duplicate checks.	aadhaarHash	Irreversible cryptographic hash. Nobody can reverse-engineer the Aadhaar number from it.
2. Envelope Field Encryption	The 12 digits are encrypted with AES-256-GCM using a customer-managed key (CMK). GCM provides authenticated encryption, instantly detecting ciphertext tampering.	aadhaarEncrypted	Only decryptable via air-gapped administrative script for statutory government audits.
3. Display Masking	Only the last 4 digits are kept in plain text (e.g. XXXX-XXXX-5678).	aadhaarLast4	Displayed on public receipts, hall tickets, and admin analytics tables.
4. Zero-Log Sanitization Middleware	Next.js API middleware intercepts all outgoing logs (CloudWatch, Sentry, Azure Monitor) and strips any 12-digit sequence matching \b\d{4}[-]?d{4}[-]?d{4}\b .	None	Guarantees zero accidental PII leakage in debug traces or crash reports.

4.2 Input Ingestion & Normalization Snippet

```
import { createHmac } from "crypto";

const normalise = (a: string) => a.replace(/\D/g, ""); // extract digits only

export function deriveAadhaarFields(rawAadhaar: string, pepper: string, kmsKey: string) {
  const clean = normalise(rawAadhaar);
  // 1. Blind Hash Index for uniqueness check (O(1) duplicate prevention)
  const aadhaarHash = createHmac("sha256", pepper).update(clean).digest("hex");
  // 2. Field-Level AES-256-GCM Ciphertext via Key Management Service
  const aadhaarEncrypted = encryptWithKms(clean, kmsKey);
  // 3. Safe Display Mask for public view and hall ticket printing
  const aadhaarLast4 = clean.slice(-4);
  return { aadhaarHash, aadhaarEncrypted, aadhaarLast4 };
}
```

5. Stage 1: Registration Engine (Nov 1 – Nov 30)

The registration engine supports single-student entries on mobile phones as well as rapid bulk batch registrations conducted by school headmasters.

5.1 Registration Field Specification

Field Name	Format / Input Type	Block	Operational Purpose
aadhaarNumber	12 digits, Verhoeff checksum	Step 0	Sole immutable unique identity; encrypted into 3 fields.
whatsappNumber	10-digit Indian mobile (+91)	Step 0	Notification routing route; non-unique (shared by teacher).
fullName	Text, as per school records	Block 1	Printed on Hall Ticket and mapped to OMR scanning software.
relativeName	Father / Mother / Guardian	Block 1	Hall ticket identity verification.
gender	MALE or FEMALE	Block 1	Drives gender-segregated exam center allocation.
standard	CLASS_9 or CLASS_10	Block 1	Academic eligibility check.
district	ANANTAPUR / SRI_SATHYA_SAI	Block 2	District quota bucket (4,000 cap per district).
mandalId	Dropdown (filtered by district)	Block 2	Proximity clustering for exam center assignment.
village	Village or Town name	Block 2	Local school cohort grouping.
schoolId	Dropdown (filtered by mandal)	Block 2	Assigns school cohort (Govt, Model, ZPHS, KGBV).
futureStream	MPC, BiPC, CEC, HEC, Poly, Undecided	Block 3	Higher secondary educational interest analysis.
vocationalInterest	Electrical, Painting, Plumbing, Carpentry, None	Block 3	Aligned with central and AP state skill missions (PMKVY/APSSDC).

5.2 District Hard Quotas & Atomic Concurrency Control

Each district is strictly capped at **4,000 confirmed registrations** (8,000 total). A two-tier locking mechanism guarantees the cap is never exceeded:

Layer	Mechanism	Behavior
Fast Edge Gate	Redis INCR district_quota:anantapur before DB write	If counter > 4,000, rejects request at edge in under 10ms with HTTP 409 "District Quota Exhausted". Reverts via DECR on rollback.
Database Guard	Atomic single-statement conditional update	Ensures exact count inside the SQL transaction. Zero updated rows rolls back immediately.

```
BEGIN;  
  -- Atomic seat reservation: check and increment in ONE atomic statement  
  UPDATE districts  
  SET registered_count = registered_count + 1  
  WHERE name = 'ANANTAPUR' AND registered_count < quota_max  
  RETURNING registered_count; -- If 0 rows returned => ROLLBACK & return HTTP 409  
  
  -- Transition status from PENDING to COMPLETED and assign registration number  
  UPDATE students  
  SET registration_status = 'COMPLETED',  
      registration_number = 'V26-ATP-B0016',  
      completed_at = NOW()  
  WHERE id = $studentId;  
COMMIT;
```

5.3 Simplified Registration Number Format

The registration number format is engineered for clarity and ease of transcription on OMR sheets:

V26 - [DIST] - [BATCH][4-DIGIT SEQ]

Example: **V26-ATP-B0016** (Anantapur, Batch B, Candidate 0016)

Component	Allowed Values	Description
V26	Static Prefix	Identifies Vadaanya Talent Test for Year 20**26**.
[DIST]	ATP or SSS	Identifies District: ATP (Anantapur) or SSS (Sri Sathya Sai).
[BATCH]	A , B , C , D	Letter designating candidate batch (e.g. Batch A: 1–1,000; Batch B: 1,001–2,000).
[4-DIGIT SEQ]	0001 to 1000	Zero-padded 4-digit sequential integer within the batch.

5.4 School-Level Cap Control (Pending Policy Decision)

- Controlled by feature flag `ENFORCE_SCHOOL_CAP`, default `false` pending final core team alignment.
- When enabled, restricts registrations to a maximum of **20 students per school**.
- Built-in exemptions for high-density institutions: AP Model Schools, KGBVs, and AP Residential Schools.
- Stored in a dynamic database configuration table so the technical lead can toggle or adjust limits instantly without code redeployment.

5.5 Post-Registration Acknowledgment Slip

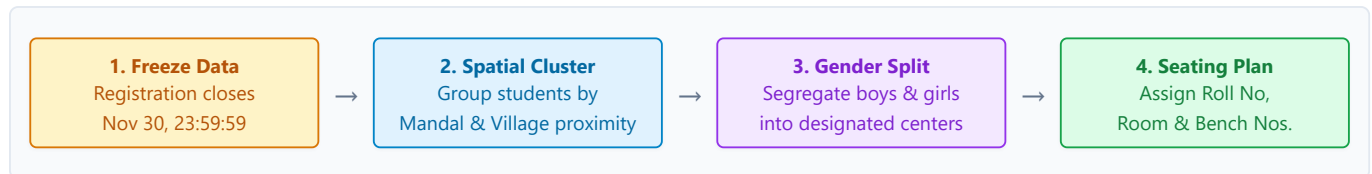
Immediately upon completion, an on-screen digital receipt is generated with:

- Student Full Name, School Name, Mandal, and District.
- Unique Registration Number: `V26-ATP-B0016`.
- High-contrast verification QR code containing encrypted student ID.
- Exam date notification: **December 15, 2026**.
- Hall ticket download availability date: **December 7, 2026**.
- One-click options to **Download Receipt PDF** or **"Quick Add Next Student"** for teachers.

6. Stages 2 to 4: Allocation, Hall Tickets and Exam Day

Hall tickets are deliberately not generated at the time of registration. Centers and seat numbers require finalized student counts across all mandals before allocation algorithms can run.

6.1 Allocation Batch Engine (Dec 1 – Dec 6)



6.2 Multi-Channel Hall Ticket Retrieval (Dec 7 – Dec 15)

Channel	Target User	Input Required	Generated Output
Direct Portal	Student / Parent	Registration Number (V26-ATP-B0016)	Single bilingual A4 PDF with QR code.
Recovery Portal	Student who lost slip	Aadhaar Number + WhatsApp Mobile	Single bilingual A4 PDF with QR code.
School Batch Download	School Headmaster / Math Teacher	School U-DISE Code + Registered Phone OTP	Single concatenated PDF of all 20–50 students in alphabetical order. Cuts printing time at rural internet cafes from 45 min to 2 min.

6.3 Hall Ticket Contents & Design

- Vadaanya Foundation official emblem and Government Education Department partnership seal.
- Student Name, Relative Name, Gender, Class, and School Name.
- Registration Number (e.g. V26-ATP-B0016) and Hall Ticket Number (e.g. HT-2026-90412).
- Assigned Examination Center name, complete physical address, and Google Maps QR/link.
- Exam schedule: **December 15, 2026**; Reporting: **09:00 AM**; Exam: **10:00 AM – 12:30 PM**.
- High-contrast 2D QR Code for rapid gate scanning on exam morning.
- Dual-language guidelines in **Telugu and English**.

6.4 Exam Day Operations (Dec 15)

- Invigilator Spot-Verification Tool** on admin.vadaanya.org : If a student misplaces their physical hall ticket on exam morning, the center headmaster searches by Name, School, or Aadhaar Last-4 to immediately verify their roll number and seat allocation.
- OMR Scanner Integration**: High-speed optical mark recognition scanners ingest candidate rosters exported in pre-formatted Excel order.

7. One-Way Automated Messaging: WhatsApp and SMS

Rural communications must route through mobile numbers rather than email. A single mobile number may be associated with multiple students, requiring personalized template messages.

7.1 Architecture & Setup

- **Primary Gateway:** Meta WhatsApp Cloud API via Next.js Server Actions / AWS Lambda.
- **Secondary Gateway:** DLT-compliant Indian transactional SMS (AWS SNS / Kaleyra / Gupshup) triggered automatically if WhatsApp delivery fails within 5 minutes.
- **Verified Profile:** Official Business Account displaying "**Vadaanya NGO**" with verified green badge.
- **Strict One-Way Enforcement:** Incoming messages trigger an automated canned auto-responder:

"This is an automated notification service from Vadaanya Foundation. Incoming messages are not monitored. For assistance, please contact your school headmaster or visit www.vadaanya.org."

7.2 The Three Notification Touchpoints

#	Trigger Timeline	Notification Purpose	Message Template Content
1	Immediately upon registration	Registration Confirmation	Dear {StudentName}, your registration for Vadaanya Talent Test 2026 is confirmed! Reg No: {RegNo}. School: {SchoolName}. Hall tickets will be available on Dec 7 at www.vadaanya.org . - Vadaanya Foundation
2	December 7, 2026	Hall Ticket Release	Dear {StudentName}, your Hall Ticket for Vadaanya Talent Test 2026 is ready! Exam Date: Dec 15, 2026. Center: {CenterName}. Download now: {ShortLink} - Vadaanya Foundation
3	December 13, 2026 (48 hrs prior)	Exam Countdown Reminder	Reminder: Vadaanya Talent Test is on Dec 15 at 9:00 AM. Center: {CenterName}. Carry your printed Hall Ticket & pen. All the best! - Vadaanya Foundation

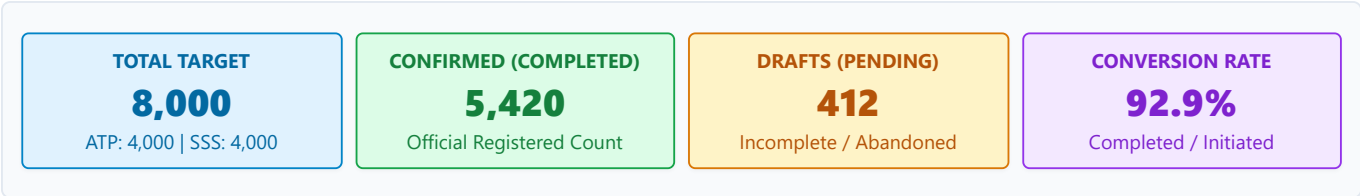
7.3 Cost Breakdown (8,000 Students × 3 Notifications)

Parameter	Projected Value
Total Student Volume × Touchpoints	8,000 students × 3 messages = 24,000 notifications
WhatsApp Utility Template Conversation Rate	≈ ₹0.12 to ₹0.15 per delivered conversation
DLT Transactional SMS Rate (India)	≈ ₹0.12 to ₹0.18 per SMS
Total Projected Messaging Expense	≈ ₹3,600 to ₹4,800 INR total

8. Admin Analytics Dashboard (admin.vadaanya.org)

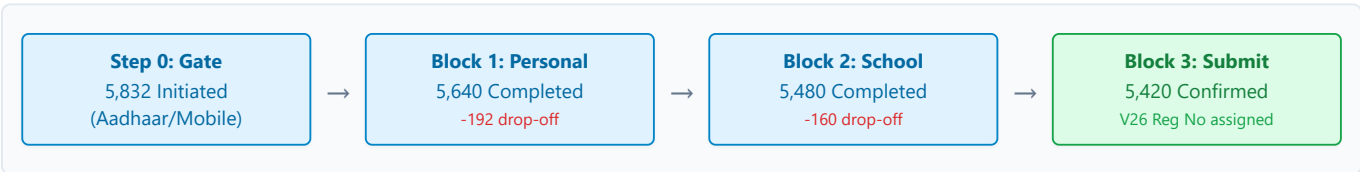
The administrative subdomain is designed **exclusively for real-time analytics, monitoring, funnel drop-off detection, and proactive field outreach**. It does not share public web traffic.

8.1 Real-Time Analytics Views & KPIs



8.2 Funnel Drop-Off Tracker

Measures exact student progression across registration steps to identify technical or logistical bottlenecks:



8.3 Hierarchical Geographic Breakdown & Click-to-Call

Hierarchy Level	Entities	Analytics Metrics Displayed	Coordinator Action Tools
State Level	Overall Program	Total 8,000 cap, completed vs pending, daily registration velocity.	Global CSV/Excel export, school cap policy switch.
District Level	Anantapur (4,000) Sri Sathya Sai (4,000)	Live progress bar toward 4,000 limit, gender ratio, class 9 vs 10 split.	District-filtered OMR export, quota override controls.
Mandal Level	64 Mandals (32 per district)	Completed registrations per mandal, active schools vs inactive schools.	Mandal summary sheet download for regional coordinators.
School Level	384+ Rural Schools	School name, category (ZPHS/Model/KGBV), completed count, pending count. Red flag if <5 registrations after Nov 10.	Click-to-Call (tel:+91...): Direct phone link to dial the Headmaster or Math Teacher with one tap.

9. Reports and Data Export Engine (.xlsx)

The administrative dashboard includes an enterprise-grade reporting engine that filters registration records and downloads structured Microsoft Excel (.xlsx) spreadsheets for operational follow-up and automated OMR evaluation.

9.1 Multi-Dimensional Filter Controls

District: All / Anantapur / Sri Sathya Sai	Mandal: Dynamic dropdown (64 Mandals)	School Category: Govt / ZPHS / Model / KGBV	Registration Status: All / COMPLETED / PENDING
Class: All / Class 9 / Class 10	Gender: All / Boys / Girls	Hall Ticket Status: All / Downloaded / Pending	Action: [Download Filtered Excel]

9.2 Specialized Excel Export Profiles

Export Profile	Target Recipient	Column Structure	Operational Value
OMR Scanner Master Export	OMR Evaluation Team	Roll_No Reg_No Full_Name Father_Name Gender Class Center_Code Room_No Bench_No School_Name	Strictly formatted column mapping for high-speed optical mark recognition scanner software.
Mandal Coordinator Outreach Sheet	Regional Coordinators	District Mandal School_Name Category Completed_Count Pending_Count HM_Name HM_Mobile_Number	Highlights lagging schools (<5 count) for field visits and immediate telephone follow-up.
Incomplete Draft Recovery Sheet	Field Volunteer Team	Aadhaar_Last4 WhatsApp_Number Full_Name District Mandal Last_Completed_Block Started_At	Allows coordinators to send recovery prompts to students who abandoned their application.
Vocational & Skills Alignment Report	APSSDC / PMKVY Liaison	Mandal School Class Future_Stream Vocational_Trade_Choice Student_Count	Aggregates student interest in Electrical, Plumbing, Carpentry, and Painting for skill initiatives.

10. Database Architecture & Complete Prisma Schema

The relational schema in PostgreSQL uses Prisma ORM to support draft saving (PENDING vs COMPLETED), field-level encryption, atomic district seat counters, and click-to-call phone tracking.

```
datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

enum RegistrationStatus {
  PENDING    // Incomplete draft
  COMPLETED // Confirmed; counts to cap
}

enum Gender {
  MALE
  FEMALE
}

enum Standard {
  CLASS_9
  CLASS_10
}

enum DistrictName {
  ANANTAPUR
  SRI_SATHYA_SAI
}

enum VocationalChoice {
  ELECTRICAL
  PAINTING
  PLUMBING
  CARPENTRY
  NOT_INTERESTED
}

model District {
  id          String    @id @default(cuid())
  name        DistrictName @unique
  quotaMax    Int        @default(4000)
  registeredCount Int    @default(0)
  mandals     Mandal[]
  students     Student[]
  createdAt   DateTime   @default(now())
}
```

```
model Mandal {
  id          String    @id @default(cuid())
  name        String
  districtId  String
  district    District  @relation(fields: [districtId],
  references: [id])
  schools      School[]
  students     Student[]
  centers       ExamCenter[]

  @@unique([name, districtId])
}

model School {
  id          String    @id @default(cuid())
  udiseCode   String?   @unique
  name        String
  category    String    // Govt, Model, ZPHS, KGBV
  mandalId    String
  mandal      Mandal    @relation(fields: [mandalId],
  references: [id])
  contactName String?   // Headmaster / Math Teacher
  contactPhone String?  // Click-to-call number
  students     Student[]
  createdAt    DateTime @default(now())

  @@index([mandalId])
}

model ExamCenter {
  id          String    @id @default(cuid())
  centerCode  String    @unique
  name        String
  address     String
  mapsUrl     String?
  mandalId    String
  mandal      Mandal    @relation(fields: [mandalId],
  references: [id])
  capacity    Int
  genderAllowed Gender? // NULL = Co-ed
  hallTickets HallTicket[]
}
```

10.1 Student, HallTicket & AuditLog Models

```
model Student {
  id String @id @default(cuid())
  registrationStatus RegistrationStatus @default(PENDING)
  currentStep Int @default(1)
  registrationNumber String? @unique // V26-ATP-B0016
  // Encrypted Security Fields
  aadhaarHash String @unique // HMAC blind
  index
  aadhaarEncrypted String // AES-256-GCM
  via KMS
  aadhaarLast4 String // Masked string
  (5678)
  whatsappNumber String // Shared (1:N)
  // Block 1: Personal Details (nullable until entered)
  fullName String?
  relativeName String?
  gender Gender?
  standard Standard?
  // Block 2: Location Details (nullable until entered)
  districtId String?
  district District? @relation(fields:
[districtId], references: [id])
  mandalId String?
  mandal Mandal? @relation(fields:
[mandalId], references: [id])
  village String?
  schoolId String?
  school School? @relation(fields:
[schoolId], references: [id])
  // Block 3: Aspirations & Skills
  futureStream String?
  vocationalInterest VocationalChoice?
  hallTicket HallTicket?
  completedAt DateTime?
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt

  @@index([districtId, registrationStatus])
  @@index([schoolId])
  @@index([whatsappNumber])
}
```

```
model HallTicket {
  id String @id @default(cuid())
  ticketNumber String @unique // HT-2026-90412
  studentId String @unique
  student Student @relation(fields:
[studentId], references: [id], onDelete: Cascade)
  centerId String
  center ExamCenter @relation(fields:
[centerId], references: [id])
  roomNumber String?
  benchNumber String?
  isDownloaded Boolean @default(false)
  downloadedAt DateTime?
  downloadCount Int @default(0)
  createdAt DateTime @default(now())

  @@index([centerId])
}

model AdminAuditLog {
  id String @id @default(cuid())
  actor String // Admin email
  action String // EXPORT_EXCEL, CAP_TOGGLED
  details String?
  ipAddress String?
  createdAt DateTime @default(now())
}
```

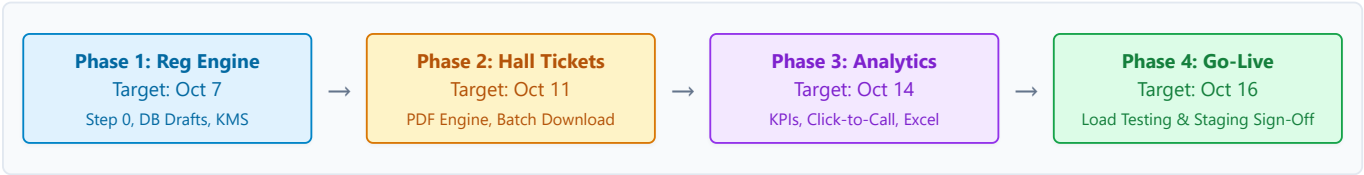
Design Note on Uniqueness: Notice that `whatsappNumber` is indexed but *not unique*. `aadhaarHash` is strictly unique. This allows teachers to register 35 students while preventing duplicate registrations.

11. Operational Edge Cases & Mitigations

#	Operational Situation	Failure Without Architectural Plan	Vadaanya 2026 System Mitigation
1	Power cut or network drop midway through form	Typed details are lost; rural student or teacher abandons application.	Database-Backed Drafts: As each block finishes, data is committed to PostgreSQL under <code>PENDING</code> . Re-entering Aadhaar restores all fields instantly.
2	Teacher registers 35 students using one phone	Phone number collision blocks student entry.	Phone number is strictly a delivery channel; Aadhaar is the unique key. "Quick Add" keeps school and mandal selections intact.
3	Student attempts duplicate registration	Duplicate records pollute database and steal quota seats.	Step 0 Gate Check: System checks <code>aadhaarHash</code> immediately and displays: "You have already registered. Reg No: V26-ATP-B0016".
4	50 simultaneous submissions at 3,995 district count	Quota overshoots 4,000 due to race conditions.	Redis atomic counter + single-statement SQL conditional update: <code>UPDATE ... WHERE registered_count < 4000</code> . Request #4,001 receives HTTP 409.
5	Student misplaces Hall Ticket on exam morning	Student is turned away at the exam hall gate.	Invigilator Spot-Verification Tool: Center headmaster searches student name or school on <code>admin.vadaanya.org</code> to confirm roll number and seat.
6	Student forgets Registration Number	Cannot download Hall Ticket.	Dual-Key Recovery: Download portal accepts <code>Aadhaar Number + Mobile Number</code> as an alternate lookup.
7	Schools with low sign-ups (<5 by Nov 10)	Unequal participation across mandals.	Automated red flag on admin analytics; coordinators tap Click-to-Call to contact headmasters directly.

12. Build Roadmap & Engineering Checklist (Oct 4 – Oct 16)

Four engineering phases take the platform from database provisioning to full production readiness by October 16, well ahead of the November 1 registration launch.



Build Phase & Target Date	Engineering Checklist Items
Phase 1: Secure Reg Engine Target: Oct 7, 2026	• Provision PostgreSQL with Prisma schema (<code>PENDING</code> vs <code>COMPLETED</code>). • Configure KMS Customer-Managed Key for AES-256-GCM encryption & HMAC blind index. • Implement Step 0 entry gate (Aadhaar & Mobile check, duplicate detection, draft resume). • Build progressive 3-block form with asynchronous DB auto-saving (<code>PATCH /api/registration/draft</code>). • Implement atomic 4,000-per-district quota enforcement with atomic SQL increment. • Implement simplified registration number format: <code>V26-[DIST]-[BATCH][SEQ]</code> . • Configure one-way WhatsApp & SMS confirmation dispatcher.
Phase 2: Hall Tickets Target: Oct 11, 2026	• Design bilingual (Telugu & English) A4 Hall Ticket layout with QR code. • Build serverless dynamic PDF generation engine. • Build individual student credential retrieval portal (<code>/talent-test/hall-ticket</code>). • Build Headmaster "School Batch Concatenated PDF" download portal. • Implement download tracking telemetry (<code>isDownloaded</code> , <code>downloadedAt</code>).
Phase 3: Analytics & Reports Target: Oct 14, 2026	• Configure <code>admin.vadaanya.org</code> subdomain with role-based access control. • Build real-time analytics dashboard: district progress bars, funnel drop-off tracker, and pending drafts counter. • Build low-registration visual alerting for schools with fewer than 5 students. • Implement click-to-call (<code>tel:+91...</code>) links for all school Headmasters. • Build Advanced Report Export engine (.xlsx) for OMR scanners and field outreach lists.
Phase 4: Hardening & Go-Live Target: Oct 16, 2026	• High-concurrency load testing (simulating 500 concurrent submissions). • Automated security verification: confirm zero plain-text Aadhaar leaks in network logs or error traces. • End-to-end rehearsal with the 50-member ground volunteer leads. • Final sign-off for public portal launch on November 1, 2026.